SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –** Hackathon	**Assessment:**	**Assessment Tool 3–** Hackathon
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	To understand the concept of supervised and unsupervised methods in machine learning.		
Student Name:	M. Poojitha Reddy	**Reg. No.:**	192472352
Section / Batch:	CSE(AI)	**Date:**	31/08/2026

Code of Conduct

I M. Poojitha Reedy/192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Poojitha Reddy

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

City/Region Grouping by Socioeconomic Indicators

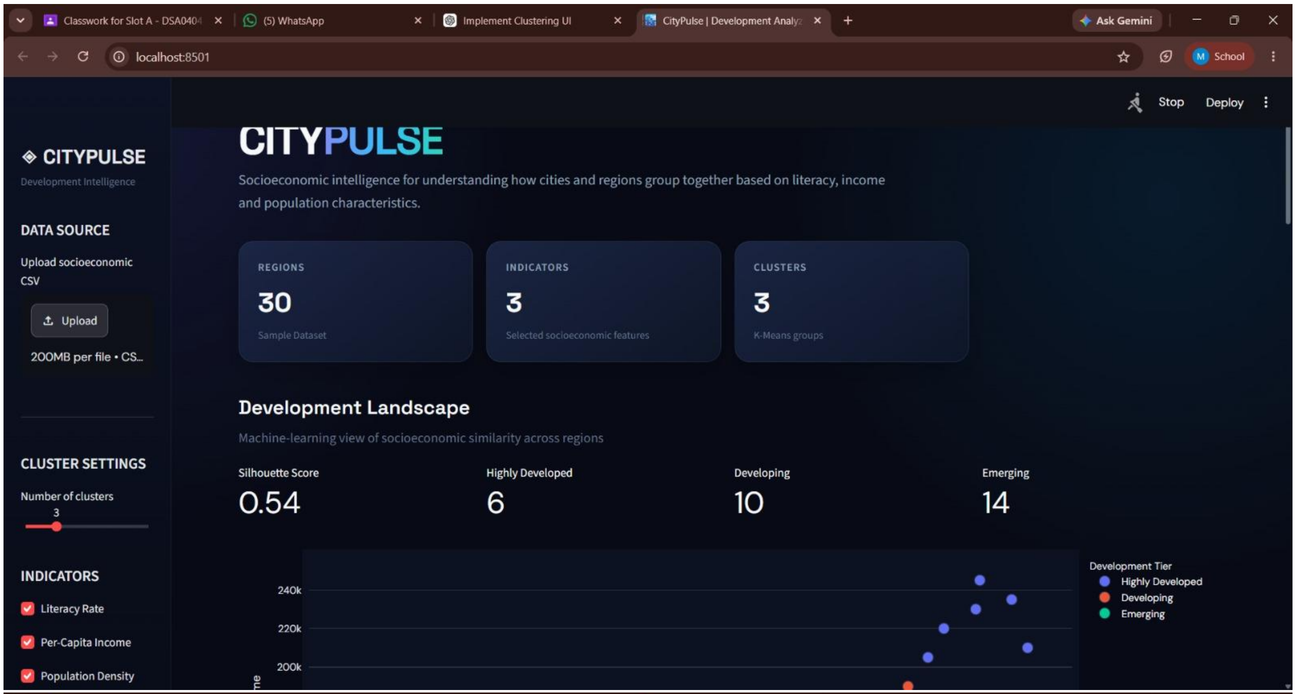
PROBLEM STATEMENT:

Socioeconomic development varies across cities and districts, making it difficult to identify regions with similar levels of development using individual indicators alone. Traditional analysis often examines factors such as literacy rate, per-capita income, and population density separately, which may not reveal the overall socioeconomic patterns among regions.

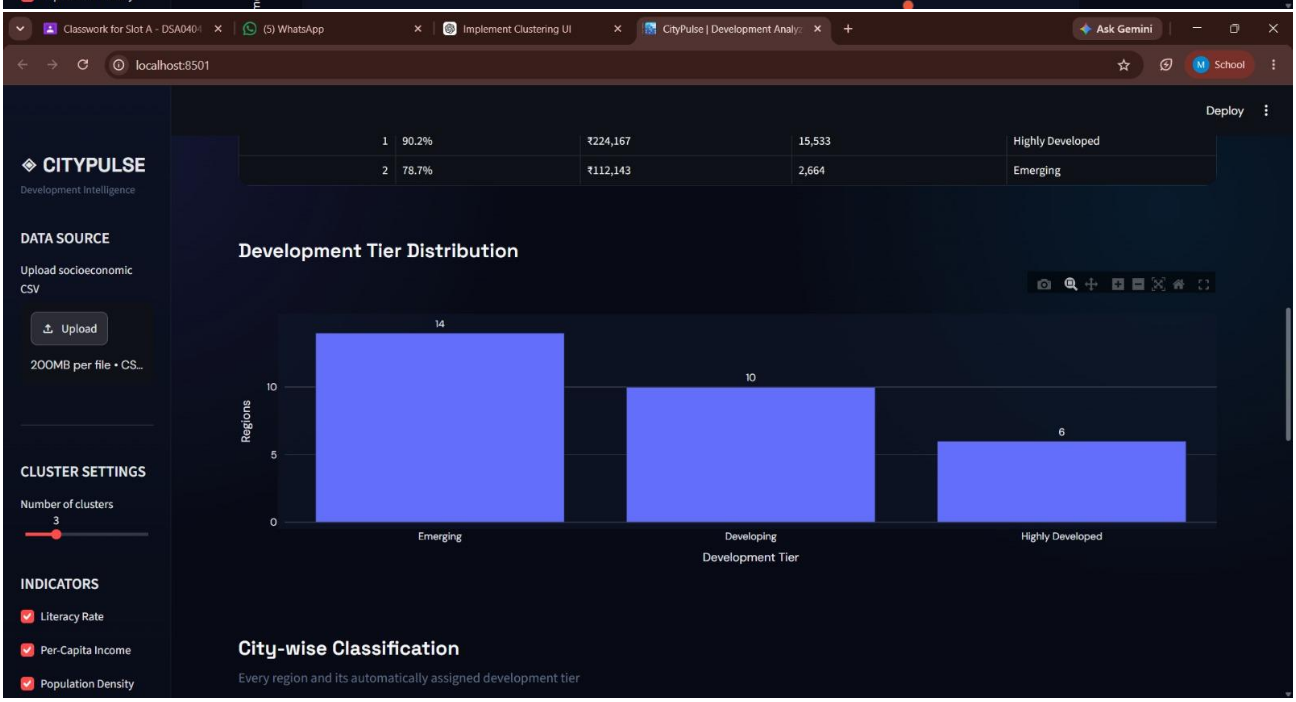
Therefore, there is a need for a *data-driven system that can automatically group cities or districts based on their socioeconomic similarities. The proposed system uses **K-Means clustering* to identify distinct regional groups and evaluates the quality of these groups using the *Silhouette Score. The clusters are then interpreted and labeled as **Highly Developed, Developing, or Emerging* based on their socioeconomic characteristics.

The system also provides an *interactive Streamlit interface* through which users can upload datasets, perform clustering, visualize results, compare socioeconomic profiles, and download the classified regional data.

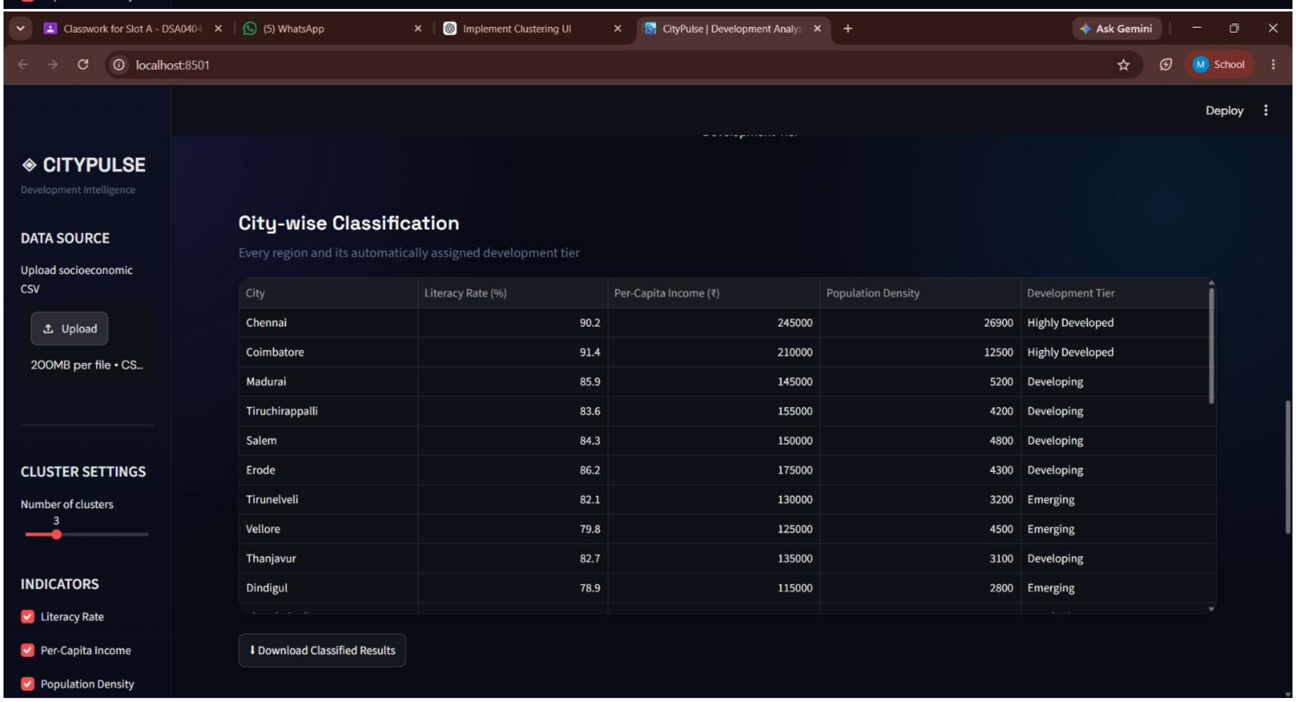
OUTPUT:



Classwork for Slot A - DSA040
(5) WhatsApp
Implement Clustering UI
CityPulse | Development Analy
Ask Gemini
localhost:8501
School
Stop
Deploy
CITYPULSE
Development Intelligence
DATA SOURCE
Upload socioeconomic CSV
Upload
200MB per file • CS...
CLUSTER SETTINGS
Number of clusters
3
INDICATORS
Literacy Rate
Per-Capita Income
Population Density
CITYPULSE
Socioeconomic intelligence for understanding how cities and regions group together based on literacy, income and population characteristics.
REGIONS
30
Sample Dataset
INDICATORS
3
Selected socioeconomic features
CLUSTERS
3
K-Means groups
Development Landscape
Machine-learning view of socioeconomic similarity across regions
Silhouette Score
0.54
Highly Developed
6
Developing
10
Emerging
14
240k
220k
200k
Development Tier
Highly Developed
Developing
Emerging

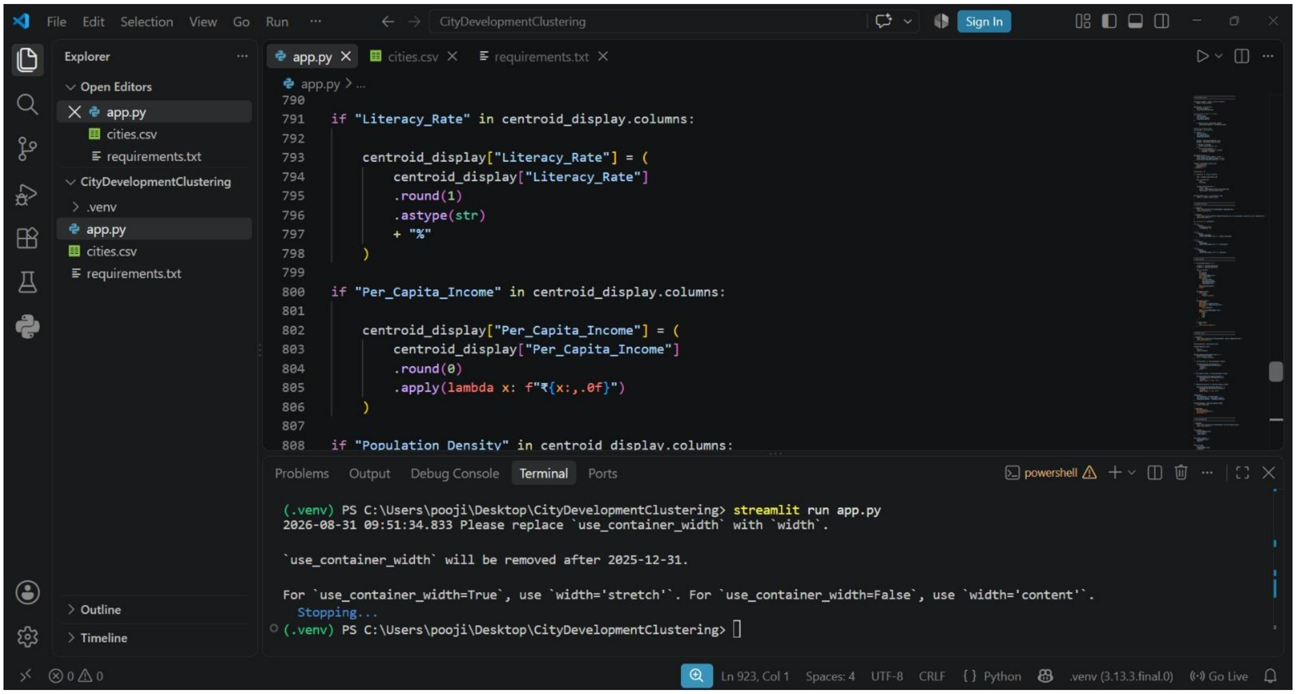


Classwork for Slot A - DSA040
(5) WhatsApp
Implement Clustering UI
CityPulse | Development Analy
Ask Gemini
localhost:8501
School
Deploy
1 90.2% ₹224,167 15,533 Highly Developed
2 78.7% ₹112,143 2,664 Emerging
CITYPULSE
Development Intelligence
DATA SOURCE
Upload socioeconomic CSV
Upload
200MB per file • CS...
CLUSTER SETTINGS
Number of clusters
3
INDICATORS
Literacy Rate
Per-Capita Income
Population Density
Development Tier Distribution
14
10
6
Regions
10
5
0
Emerging
Developing
Highly Developed
Development Tier
City-wise Classification
Every region and its automatically assigned development tier

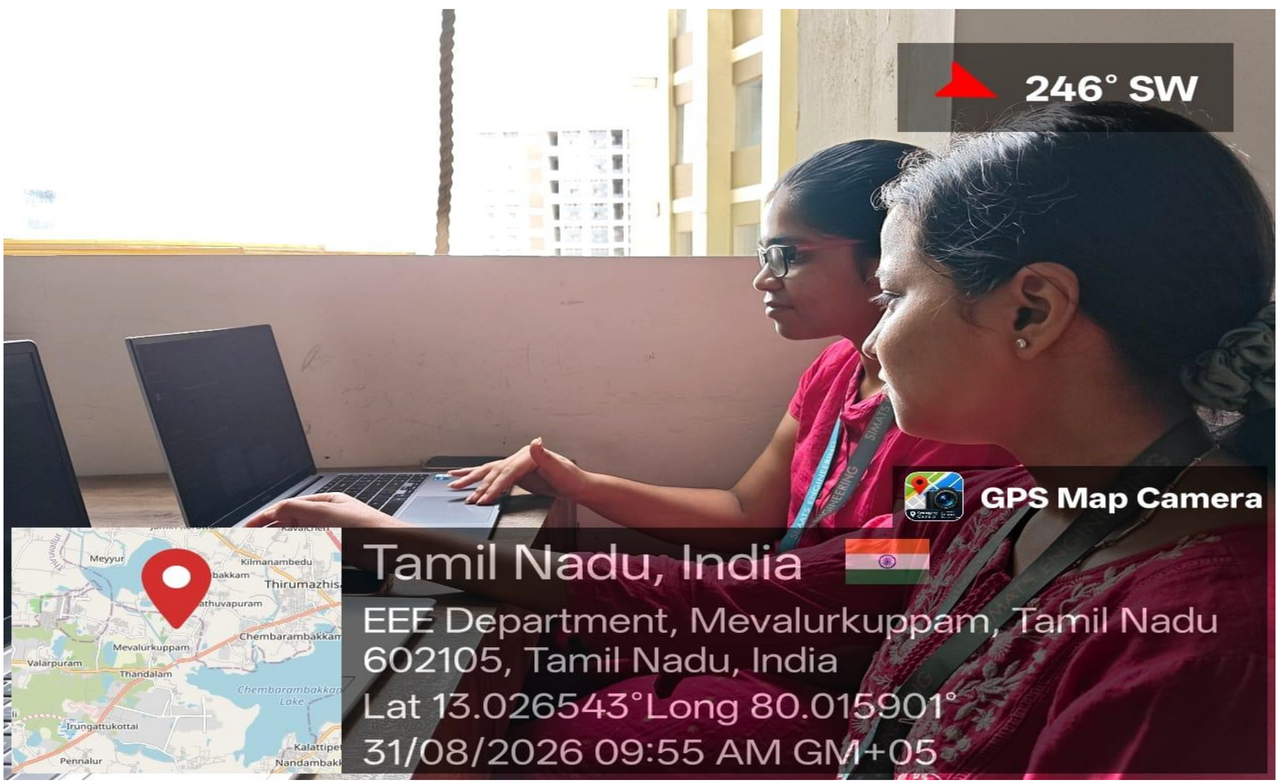


City	Literacy Rate (%)	Per-Capita Income (₹)	Population Density	Development Tier
Chennai	90.2	245000	26900	Highly Developed
Coimbatore	91.4	210000	12500	Highly Developed
Madurai	85.9	145000	5200	Developing
Tiruchirappalli	83.6	155000	4200	Developing
Salem	84.3	150000	4800	Developing
Erode	86.2	175000	4300	Developing
Tirunelveli	82.1	130000	3200	Emerging
Vellore	79.8	125000	4500	Emerging
Thanjavur	82.7	135000	3100	Developing
Dindigul	78.9	115000	2800	Emerging

CODE:



File Edit Selection View Go Run
CityDevelopmentClustering
Sign In
Explorer
Open Editors
app.py
cities.csv
requirements.txt
CityDevelopmentClustering
.venv
app.py
cities.csv
requirements.txt
if "Literacy_Rate" in centroid_display.columns:
centroid_display["Literacy_Rate"] = (
centroid_display["Literacy_Rate"]
.round(1)
.astype(str)
+ "%"
)
if "Per_Capita_Income" in centroid_display.columns:
centroid_display["Per_Capita_Income"] = (
centroid_display["Per_Capita_Income"]
.round(0)
.apply(lambda x: f"₹{x:,.0f}")
)
if "Population Density" in centroid display.columns:
Problems Output Debug Console Terminal Ports
powershell
(.venv) PS C:\Users\pooji\Desktop\CityDevelopmentClustering> streamlit run app.py
2026-08-31 09:51:34.833 Please replace `use_container_width` with `width`.
`use_container_width` will be removed after 2025-12-31.
For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
Stopping...
(.venv) PS C:\Users\pooji\Desktop\CityDevelopmentClustering>
Outline
Timeline
Ln 923, Col 1 Spaces: 4 UTF-8 CRLF {} Python .venv (3.13.3.final.0) Go Live



246° SW
GPS Map Camera
Tamil Nadu, India
EEE Department, Mevalurkuppam, Tamil Nadu
602105, Tamil Nadu, India
Lat 13.026543° Long 80.015901°
31/08/2026 09:55 AM GMT+05
Meyyur
Kilmanambedu
Thirumazhisai
Mevalurkuppam
Chembarambakkam
Valarpuram
Thandalam
Chembarambakkam Lake
Irungattukottai
Kalattipet
Pennalur
Nandambakkam